
Theoretical Derivations from Lecture Material

1. Hydrostatic Equilibrium Starting from a small mass element ΔM in a thin spherical shell at radius r , explain how the balance between the inward gravitational force and the outward pressure gradient force leads to the hydrostatic equilibrium equation

$$\frac{dP}{dr} = -\rho(r)g(r) \quad (1)$$

Hint: In your derivation, clearly show all of your assumptions, such as how you evaluated $P(r)$ and $P(r + \Delta r)$. State the physical significance of the negative sign in the final expression

2. Free-Fall Time (t_{ff}) Starting from the energy conservation equation for a spherical shell of mass at radius r given by $\frac{1}{2} \left(\frac{dr}{dt} \right)^2 = \frac{Gm_0}{r} - \frac{Gm_0}{r_0}$ where the star started from r_0 and freely collapses to r , derive the characteristic free-fall time-scale expression

$$t_{ff} = \sqrt{\frac{3\pi}{32G\rho}} \quad (2)$$

Hint: In your derivation outline, clearly describe the separation of variables step, the substitution used to simplify the integral and how the limits of integration are chosen from $r = r_0$ to $r = 0$. Additionally, explain why the initial velocity is assumed to be zero at $r = r_0$, and what physical assumption about the system this represents

3. Virial Theorem Starting from the hydrostatic equilibrium equation, multiply both sides by the spherical volume element $4\pi r^3$ and integrate from the center of the star ($r = 0$) to its surface ($r = R$). Using integration by parts, show that the volume-averaged pressure is related to the gravitational potential energy through the relation

$$-3\langle P \rangle V = E_{GR} \quad (3)$$

Hint: In your derivation, clearly outline each mathematical step and explain how integration by parts is applied to the pressure term. Also, state the physical assumptions at the center ($r = 0$) and at the surface ($r = R$) that justify setting the boundary term $[P(r) 4\pi r^3]_0^R$ to zero and explain what these conditions imply about pressure behavior in a self-gravitating spherical system

4. Relativistic Pressure Regimes Starting from the kinetic theory of gases where pressure is defined as $P = \frac{1}{3}n\langle p \cdot v \rangle$, provide a step-by-step derivation to show that for a gas composed of non-relativistic particles ($p = mv$), the pressure is exactly $P = \frac{2}{3} \frac{E_{KE}}{V}$. Once you have established this pressure relation, substitute it into the volume-integrated virial relation $-3\langle P \rangle V = E_{GR}$ to arrive at the non-relativistic Virial Theorem

$$2E_{KE} + E_{GR} = 0 \quad (4)$$

5. Stellar Thermostat Mathematically demonstrate that when a star loses energy to space it must shrink and get hotter rather than cooling down. Why is this strange negative heat capacity necessary for a young star to eventually get hot enough to start nuclear fusion?

Theoretical Derivations to Numerical Validation

6. Estimating Core Pressure A typical young stellar object (YSO) has a mass $M = 2.0 \times 10^{30}$ kg and a radius $R = 7.0 \times 10^8$ m. Assuming the star has a uniform density and is in hydrostatic equilibrium use virial relation to estimate the volume-averaged internal pressure $\langle P \rangle$ of the star

7. The Speed of Collapse A giant molecular cloud (GMC) core has an average mass density of $\rho = 2 \times 10^{-16} \text{ kg/m}^3$. Calculate the free-fall time-scale t_{ff} in years. If a second cloud is 10 times more massive but has the same density, how does its collapse time change?

8. Thermal Energy of a Protostar Consider a protostar that has contracted until its gravitational potential energy is $E_{GR} = -4.0 \times 10^{41} \text{ J}$. Assuming the gas is non-relativistic, use the Virial Theorem ($2E_{KE} + E_{GR} = 0$) to 1) find the total internal kinetic energy E_{KE} . 2) Calculate the total energy E_{tot} of the system. If the star radiates 10^{35} J of energy away, calculate the new E_{KE} and explain if the temperature increased or decreased

9. Relativistic Transitions At the end of a massive star's life, the core density becomes so high that electrons become ultra-relativistic. If the internal energy density remains $\frac{E_{KE}}{V} = 10^{24} \text{ J/m}^3$, calculate the pressure P for the particles in Non-Relativistic and the particles in Ultra-Relativistic ($P = \frac{1}{3} \frac{E_{KE}}{V}$) regimes. Briefly explain which state provides more support against gravitational collapse

10. The Solar Limit The Sun is currently in hydrostatic equilibrium with an average density of $\rho_{\odot} \approx 1400 \text{ kg/m}^3$. If the Sun's internal pressure were suddenly "turned off," calculate the time (in minutes) it would take for the Sun to collapse into a point mass. If the Sun were 8 times more massive but occupied the same volume, would the collapse happen faster or slower? Explain using the relationship between ρ and t_{ff}

11. Supernova Limits In the final moments of a massive star, the iron core ($\rho_{core} \approx 10^{13} \text{ kg/m}^3$) becomes unstable and collapses. Calculate the free-fall time (t_{ff}) for this core collapse in milliseconds. Assume that transparent clouds collapse on the t_{ff} scale while opaque clouds slow down. During a supernova, the core is extremely dense and opaque. Why, then, does it still collapse at nearly the free-fall speed?

————— Theoretical Derivations to Astrophysical Observations —————

12. Speed Limits An astronomer observes that the centers of dense molecular cores are much brighter and more compact than their outer shells. Using the free fall time scale explain why an observer sees this pattern, and why the center reaches the high-density stardom phase significantly earlier than the outer edge

13. The Infrared Protostar Protostars are often called infrared sources because they are not yet hot enough for hydrogen fusion but are still glowing. According to the Virial Theorem ($2E_{KE} + E_{GR} = 0$), where does the energy for this infrared luminosity come from if the star has not yet started its stardom fusion phase?

14. The Brown Dwarf Limit An Astronomer observes many failed stars (Brown Dwarfs) that never reach the main sequence. Based on the Stellar Thermostat concept explain why does an object with very low mass ($M < 0.08M_{\odot}$) fail to reach the 10^7 K required for fusion, even though it is contracting and releasing gravitational potential energy?

15. Massive Stars Massive stars ($M > 20M_{\odot}$) have much higher surface temperatures than the Sun. Using the Virial Theorem relation $\langle P \rangle = -\frac{1}{3} \frac{E_{GR}}{V}$, explain why a more massive star must maintain a much higher internal temperature to stay in equilibrium

16. Supernova Triggers An astronomer accidentally observes a massive star reaches the end of its life becoming a core-collapse supernova. If the core's particles become ultra-relativistic how does the shift from $P \propto \frac{2}{3} E_{density}$ to $P \propto \frac{1}{3} E_{density}$ explain why the star can no longer support itself against gravity, leading to a visible explosion?